create.project: One-Call Reproducible Research Projects, R Packages and Papers*

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Abstract

Setting up an R project properly takes time: folders, git, pinned package versions, a license, a paper. Most people skip it and pay for it later. `create.project` does it in one call. `create_project()` builds either a research project (a numbered data pipeline with `renv`) or an R package (the standard layout, with tests), and makes the first git commit. `create_paper()` writes a LaTeX paper skeleton, for research or for software, that compiles on day one. Every step skips what already exists, so re-running is safe. The package is about 300 lines of R on top of `usethis`, `renv` and `gert`, and is open source under the MIT license.

Keywords: R, reproducibility, project templates, research workflow, LaTeX

*Word count: about 1,000.

1 Statement of Need

Good practices for research code are well known: keep raw data read-only, number the scripts in execution order, pin package versions, use version control, and generate figures instead of pasting them (Marwick et al., 2018; Wilson et al., 2017). They are also slow to set up, and the setup happens at the worst moment: the start of a project, when nothing seems at stake yet. So each project gets a different layout, or none.

`create.project` makes the good default the fast default. It targets social scientists who write R, and anyone who writes small R packages. One call gives a project that a colleague, a reviewer, or the same person on another computer can rerun.

2 State of the Field

`usethis` (Wickham et al., 2026) provides the building blocks (create a package, add a license, add tests) but no research layout and no paper. `rrtools` (Marwick et al., 2018) turns a research project into a package (a "research compendium") with an R Markdown paper and Docker. `ProjectTemplate` (White, n.d.) gives a data analysis layout. `workflowr` (Blischak et al., 2019) turns R Markdown analyses into a versioned website (Table 1).

`create.project` sits between them: research and package layouts behind one function, and a LaTeX paper rather than R Markdown. Users who write in R Markdown or Quarto are better served by `rrtools` or `workflowr`.

Table 1: Comparison with existing tools

	`create.project`	`usethis`	`rrtools`	`ProjectTemplate`	`workflowr`
Research layout	✓		✓	✓	✓
Package layout	✓	✓	✓		
Paper	LaTeX		R Markdown		R Markdown
Version pinning	`renv`		Docker		

3 Design

3.1 Principles

1. **One call, sensible defaults.** Author, license and branch name are preset. The only required argument is the folder.

2. **Idempotent.** Every step skips what already exists. Re-running never overwrites work, and an existing folder can be upgraded.
3. **Delegate.** Files are created by `usethis`, environments by `renv` (Ushey & Wickham, 2026), commits by `gert` (Ooms, 2026). The package only decides *what* to create.
4. **The paper compiles on day one.** Placeholders in brackets are writing guidance; example figures and tables (Star Wars data) fill the blanks; a missing figure is drawn as a labelled box instead of breaking the build.

3.2 Architecture

Two exported functions (Figure 1). `create_project()` chooses a layout from `type`, then calls `create_paper()` with the matching paper type. `create_paper()` also works alone, in any project. Table 2 lists what each layout contains.

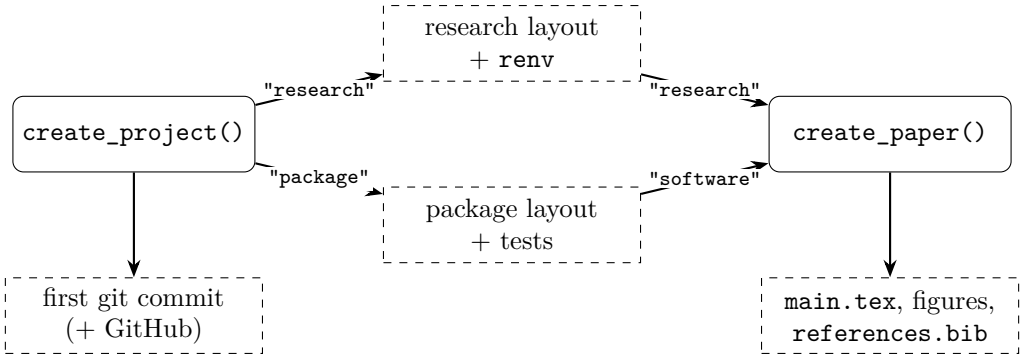


Figure 1: Architecture: two exported functions and what they produce

Table 2: The two layouts

`type = "research"`	`type = "package"`
`1_raw_data/` never edited by hand	`R/` all functions
`2_scripts/` numbered = run order	`man/` generated documentation
`3_derived_data/` regenerated	`tests/testthat/` unit tests
`4_outputs/` figures, tables	`vignettes/` tutorial
`5_docs/` research paper	`data-raw/` scripts that build `data/`
`6_archives/` old material	`inst/CITATION`
`renv/` pinned package versions	`paper/` software paper
both: `DESCRIPTION`, `.Rproj`, `LICENSE`, `README.md`, ignore files, git	

3.3 Decisions and Trade-offs

- **One function with `type`, not two functions.** About 80% of the steps are shared; two functions would drift apart.
- **`renv` for research only.** A package declares its dependencies in `DESCRIPTION`; pinning them would fight its users.
- **No options that nobody used.** Flags for git, `renv` and data splits, a generated folder map and a per-computer setup function were removed: each cost code and explanation for little benefit.
- **The repository is named after the package** (`my.project`, not `my_project`), because `usethis` does so and offers no override.
- **Example figures are drawn, not copied.** The duel in Figure 2 is made with `ggplot2`; film stills are copyrighted.

4 Usage

Install once, and keep `renv` libraries out of synced folders (OneDrive, Dropbox) with one line in the user `.Renviron`:

```
remotes::install_github("SimonLabarthe/create.project")
usethis::edit_r_environ()   # add: RENV_PATHS_LIBRARY_ROOT=<local folder>
```

Then one call per project:

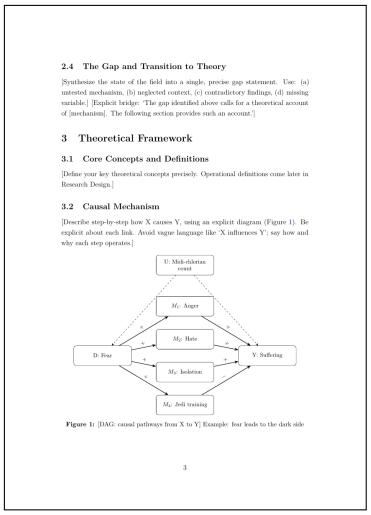
```
library(create.project)

create_project("~/seminar/inequality", title = "Inequality and Repression"
    )
create_project("~/dev/coolfun", type = "package")
create_paper("5_docs")          # only the paper, in an existing project
```

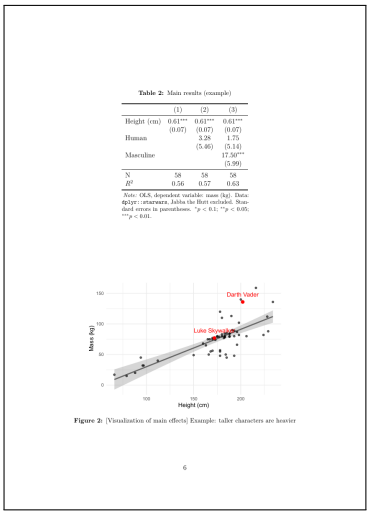
The research call writes the six folders, five numbered script templates, a paper and a first commit. The paper already contains a causal diagram, two tables and five figures (Figure 2); each is an example to replace.

5 Quality and Validation

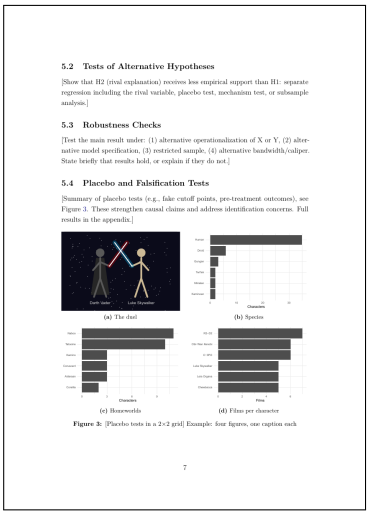
22 `testthat` expectations cover both layouts, re-running on an existing project, the paper templates and the escaping of LaTeX special characters. `R CMD check` reports no errors, warnings or notes, and so does a package freshly generated with `type =`



(a) Causal diagram



(b) Results



(c) Figure grid

Figure 2: Three pages of a freshly generated research paper, before any writing

`"package"`. Both paper types compile without LaTeX warnings with `pdflatex` and `biber`. Everything was tested on Windows 11 with R 4.5.2.

6 Limitations and Future Work

Defaults (author, GitHub account) are the author's own: other users must pass them as arguments. macOS and Linux are untested. Papers are LaTeX only. Next steps: read personal defaults from `options()`, add a continuous integration workflow, and offer a Quarto paper.

7 Conclusion

Good project structure is cheap once it is one function call away. `create.project` makes that call, and gives the paper a head start.

Software Availability

- Repository: https://github.com/SimonLabarthe/create.project
- Version described: 0.3.0
- License: MIT
- Requirements: R, and a LaTeX distribution (e.g. TinyTeX) for the papers

References

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